Neuro-Fuzzy Computing - Problem-set 3rd: Teamwork

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- Problem 01

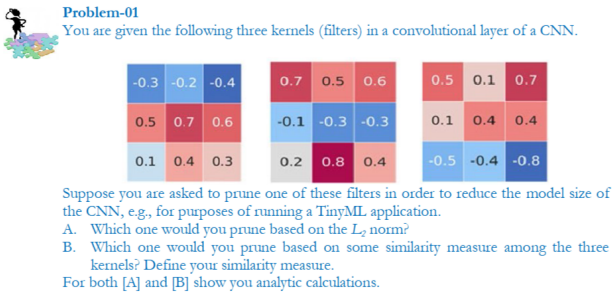


Figure 1: Problem 01

A

For the first filter :

$L_2 = \sqrt{(-0.3)^2 + (0.5)^2 + (0.1)^2 + (-0.2)^2 + (0.7)^2 + (0.4)^2 + (0.4)^2 + (0.6)^2 + (0.3)^2}$

$L_2 = \sqrt{0.09 + 0.25 + 0.01 + 0.04 + 0.49 + 0.16 + 0.16 + 0.36 + 0.09}$

$L_2 = \sqrt{1.65} = 1.2845$ For the second filter :

$L_2 = \sqrt{(0.7)^2 + (-0.1)^2 + (0.2)^2 + (0.5)^2 + (-0.3)^2 + (0.8)^2 + (0.6)^2 + (-0.3)^2 + (0.4)^2}$

$L_2 = \sqrt{0.49 + 0.25 + 0.36 + 0.01 + 0.09 + 0.09 + 0.04 + 0.64 + 0.16}$

$L_2 = \sqrt{2.13} = 1.4594$ For the third filter :

$L_2 = \sqrt{(0.5)^2 + (0.1)^2 + (0.7)^2 + (0.1)^2 + (0.4)^2 + (0.4)^2 + (-0.5)^2 + (-0.4)^2 + (0.8)^2}$

$L_2 = \sqrt{0.25 + 0.01 + 0.49 + 0.16 + 0.16 + 0.25 + 0.01 + 0.16 + 0.64}$

$L_2 = \sqrt{1.97} = 1.4035$

The smallest L_2 is in the first filter so i select that one **B**

The cosine similarity between two vectors A and B is given by:

$$\cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|} \tag{1}$$

where:

- $A \cdot B$ is the dot product.
- $\|A\|$ and $\|B\|$ are the Euclidean norms.

Flattening each matrix into vectors:

$$K_1 = [-0.3, -0.2, -0.4, 0.5, 0.7, 0.6, 0.1, 0.4, 0.3]$$
$$K_2 = [0.7, 0.5, 0.6, -0.1, -0.3, -0.3, 0.2, 0.8, 0.4]$$
$$K_3 = [0.5, 0.1, 0.7, 0.1, 0.4, 0.4, -0.5, -0.4, -0.8]$$

Dot products

$$\begin{aligned} K_1 \cdot K_2 &= (-0.3 \times 0.7) + (-0.2 \times 0.5) + (-0.4 \times 0.6) + (0.5 \times -0.1) + (0.7 \times -0.3) + (0.6 \times -0.3) + (0.1 \times 0.2 \\ &= -0.21 - 0.10 - 0.24 - 0.05 - 0.21 - 0.18 + 0.02 + 0.32 + 0.12 = -0.63 \end{aligned}$$

$$\begin{aligned} K_1 \cdot K_3 &= (-0.3 \times 0.5) + (-0.2 \times 0.1) + (-0.4 \times 0.7) + (0.5 \times 0.1) + (0.7 \times 0.4) + (0.6 \times 0.4) + (0.1 \times -0.5) + \\ &= -0.15 - 0.02 - 0.28 + 0.05 + 0.28 + 0.24 - 0.05 - 0.16 - 0.24 = -0.33 \end{aligned}$$

$$\begin{aligned} K_2 \cdot K_3 &= (0.7 \times 0.5) + (0.5 \times 0.1) + (0.6 \times 0.7) + (-0.1 \times 0.1) + (-0.3 \times 0.4) + (-0.3 \times 0.4) + (0.2 \times -0.5) + \\ &= 0.35 + 0.05 + 0.42 - 0.01 - 0.12 - 0.12 - 0.10 - 0.32 - 0.32 = -0.17 \end{aligned}$$

Norm computation

$$\|K_1\| = \sqrt{1.65} \approx 1.284, \quad \|K_2\| = \sqrt{2.13} \approx 1.459, \quad \|K_3\| = \sqrt{1.97} \approx 1.4035$$

Final Cosine Similarity Values:

$$\cos(K_1, K_2) = \frac{-0.63}{1.284 \times 1.459} \approx -0.283$$
$$\cos(K_1, K_3) = \frac{-0.33}{1.284 \times 1.403} \approx -0.176$$
$$\cos(K_2, K_3) = \frac{-0.17}{1.459 \times 1.403} \approx -0.080$$

The less negative value is between second and third, that means that we can choose one of them to be deleted.

- Problem 02

A.

-start with presenting p1:

$$\mathbf{a} = \mathbf{compet}(\mathbf{W}\mathbf{p_1}) =$$

$$= \mathbf{compet}\left(\begin{bmatrix} 0 & -1 \\ \frac{-2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix}\right)$$

$$= \mathbf{compet}\left(\begin{bmatrix} 0 \\ \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

w_2 wins the competition.

-start with presenting p2:

$$\mathbf{a} = \mathbf{compet}(\mathbf{W}\mathbf{p_2}) =$$

$$= \mathbf{compet}\left(\begin{bmatrix} 0 & -1 \\ \frac{-2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

$$= \mathbf{compet}\left(\begin{bmatrix} -1 \\ \frac{-1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

w_2 wins the competition.

-start with presenting p3:

$$\mathbf{a} = \mathbf{compet}(\mathbf{W}\mathbf{p_2}) =$$

$$= \mathbf{compet}\left(\begin{bmatrix} 0 & -1 \\ \frac{-2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}\right)$$

$$= \textbf{compet}\left(\begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{-3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

w_1 **wins the competition and moves closer to p3. So, we will start with the presentation of p3.**

B.

Problem's solution exists in **problem02-set03.ipynb**.

C.

w_1 **wins on the 14th presentation and, specifically, presentation of p3.**

- Problem 03

$NFC(x) = x * tanh(e^x) \implies$ activation function,

where $tanh(x) = \frac{sinh(x)}{cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

We will try to figure out vanishing and exploding gradient problem by observing the diagram of the activation function NFC.

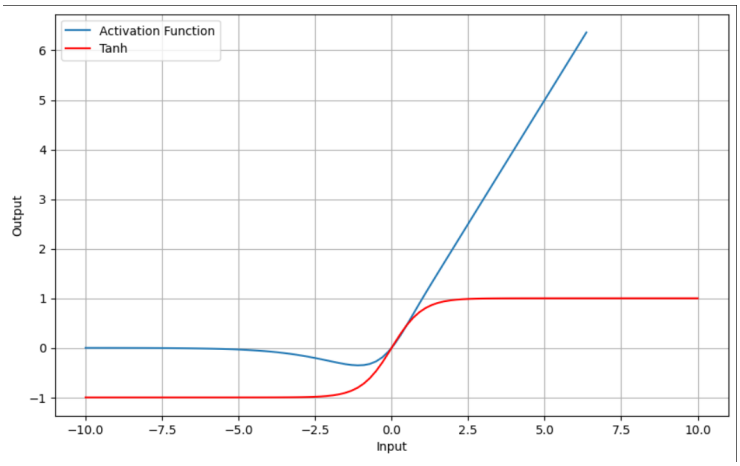


Figure 2: problem03-set03.ipynb

- Vanishing Gradient problem:

- flat region for $x < -2 \implies$ derivative in this region is close to 0. **This leads to vanishing gradients.**
- $x > -1 \implies$ output grows $\implies$ gradients non-zero $\implies$ vanishing problem does not occur here.

- Exploding Gradient problem:

- Exploding gradients grow larger instead of smaller during backpropagation.
- This phenomenon occurs when the gradients of the loss function with respect to the network's weights become excessively large.
- **Activation functions** can contribute to the problem by allowing large outputs that lead to high gradients.
- Although outputs are getting larger and larger for $x > -1$, **this does not lead to high gradients and, specifically, exploding gradient problem.** That's because for $x > -1$ there is **y=ax function, which has gradient a and is the same for every x**. So, outputs are getting larger and larger for $x > -1$ in our activation function, but as long as the form of these outputs is y=ax, the gradients will be the same (y'=a) for every x, which does not lead to exploding gradient problem.

- Problem 04

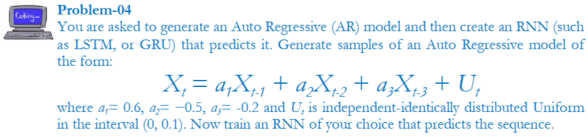


Figure 3: data

Data Generation
To create the dataset, the AR model was simulated over 5000 time steps. The first three values were initialized as zeros, and subsequent values were computed using the AR equation. The noise term U_t was sampled from a uniform distribution.

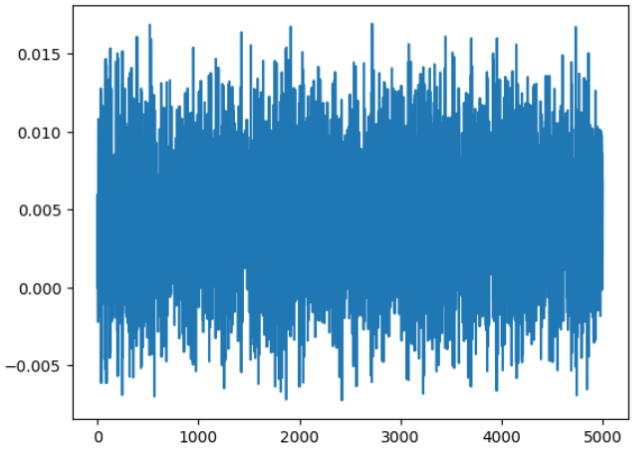


Figure 4:

Preprocessing

Before training the LSTM model, the data was structured in a supervised learning format, where each input consists of three lagged values $(X_{t-1}, X_{t-2}, X_{t-3})$, and the target is X_t. The dataset was then split into training (80%) and testing (20%). The features and targets were normalized using Min-Max scaling to enhance training stability.

LSTM Model Architecture

The LSTM model was implemented using TensorFlow/Keras. The architecture consists of:

- An LSTM layer with 25 units and ReLU activation function.
- A Dense layer with a single neuron for output prediction.
- Adam optimizer and Mean Squared Error (MSE) as the loss function.

The model was trained for 15 epochs with a batch size of 4. Different training sample sizes (500, 2000, 4000) were tested to evaluate the effect of dataset size on model performance.

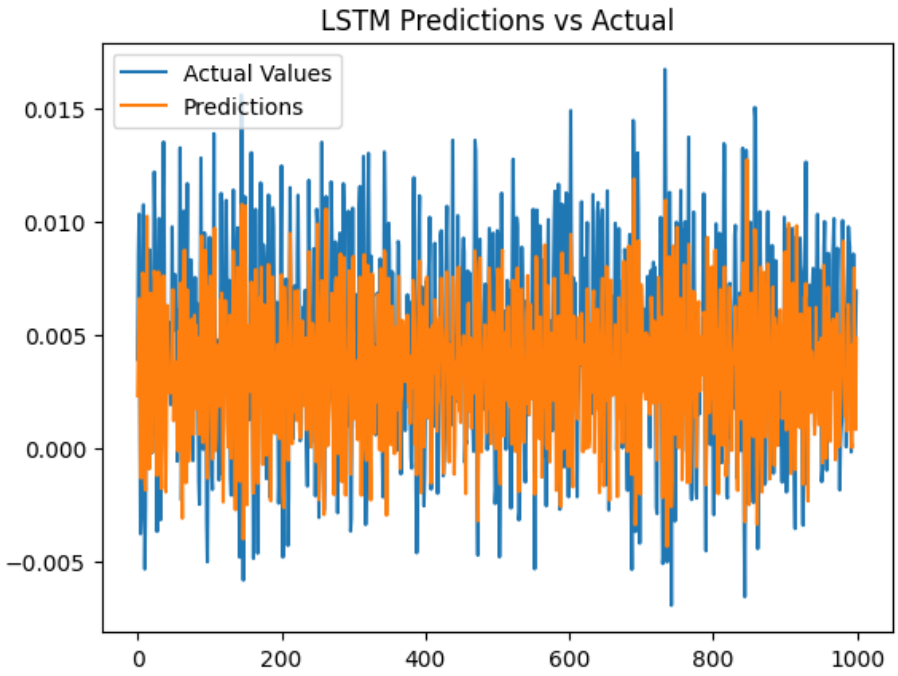


Figure 5:

0.1 Results and Discussion

The final evaluation was performed on the test set. The MSE values obtained for different training sizes were:

```
Training size: 500, MSE Loss on test set: 0.016063977032899857
Training size: 2000, MSE Loss on test set: 0.014219890348613262
Training size: 4000, MSE Loss on test set: 0.015907375141978264
MSE loss on new dataset: 0.016330674290657043
```

Figure 6: Problem 9

As expected, increasing the training size led to better generalization, reducing the test error. The LSTM model effectively captured the dependencies within the AR(3) process, showing that recurrent neural networks can successfully learn temporal dependencies in autoregressive models.

- Problem 05

- Problem 06

Fuzzy Set Operations:

1. A
   A^c: $m_{A'}(x) = 1 - m_A(x) -> A^c(x) = 1 - A(x)$
2. $A \cup B$
   $(A \cup B)(x) = max(A(x), B(x))$
3. $A \cap B$
   $(A \cap B)(x) = min(A(x), B(x))$

Solution:
a) $A \cap B = min(A(x), B(x)) = [(A|0), (B|0.3), (C|0.5), (D|0.8), (E|0), (F|0.2), (G|0.6)]$

b) $A \cup B = max(A(x), B(x)) = [(A|0.3), (B|1), (C|0.7), (D|1), (E|1), (F|0.5), (G|0.6)]$

c) $A \cap B^c = min(A, 1-B) = [(A|0), (B|0), (C|0.5), (D|0.2), (E|0), (F|0.2), (G|0.4)]$

d) $(A \cup B^c) \cap C = min(max(A, 1-B), C) = [(A|0.7), (B|0.3), (C|0.5), (D|0.2), (E|0), (F|0.2), (G|0.6)]$

e) $(A \cap B^c) \cup C^c = max(min(A, 1-B), 1-C) = [(A|0), (B|0.5), (C|0.5), (D|0.8), (E|1), (F|0.8), (G|0.4)]$

f) $(A \cap A^c) \cup A = max(min(A, 1-A), A) = [(A|0), (B|0.3), (C|0.7), (D|1), (E|0), (F|0.2), (G|0.6)]$

- Problem 07

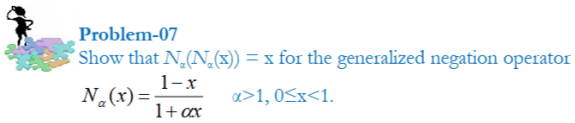


Figure 7: data

In order to solve this, we are going to put it the equaztion $N_a(x)$ over x.
Thus $N_a(x) = \frac{1-\frac{1-x}{1+ax}}{1+a\frac{1-x}{1+ax}} = \frac{\frac{1+ax-1+x}{1+ax}}{\frac{1+ax+a-ax}{1+ax}} = \frac{ax+x}{1+a} = \frac{x(a+1)}{a+1} = x$

- Problem 08

$\mu_A(x) = \frac{1}{1+x}$

Show that it **does not** satisfy the law of contradiction and the law of excluded middle.

-Law of contradiction in fuzzy expression:

Rule: min($\mu_A(x), 1 - \mu_A(x)$)=0

$\mu_A(x) * \mu_{A^c}(x) = 0,$

with $\mu_{A^c}(x) = 1 - \mu_A(x) = 1 - \frac{1}{1+x} = \frac{x+1-1}{x+1} = \frac{x}{x+1}$

No element can belong both to set A and its complement A^c.

For x=1:

- $\mu_A(x) = \frac{1}{1+x} \implies \mu_A(x) = \frac{1}{1+1} = \frac{1}{2}$
- $\mu_{A^c}(x) = \frac{x}{1+x} \implies \mu_{A^c}(x) = \frac{1}{1+1} = \frac{1}{2}$

As we can see, element x=1 belongs to both set A and set A^c. So, $\mu_A(x) = \frac{1}{1+x}$ does not satisfy the law of contradiction.

-Law of excluded middle in fuzzy expression:

Rule: max($\mu_A(x), 1 - \mu_A(x)$)=1

Theorem: $P \vee \neg P$

Either P is true, or P in not true.

For $x = 1$:

$\mu_A(1) = \frac{1}{1+1} = 0.5$

$\mu_{A^c}(x) = \frac{x}{1+x} \implies \mu_{A^c}(1) = \frac{1}{1+1} = 0.5$

$max(0.5, 0.5) = 0.5$

which means that it does not satisfy the law of excluded middle.

- Problem 09

From the exercise we have these: $t_L(A- > B) = min(1, 1 - t(A) + t(B))$ (1)
and these :
$t_L(A- > B) = c$ (2) and $t(A) \geq a$ (3)
From (1), (2) : $min(1, 1 - t(A) + t(B)) = c$ with values between 0 and 1



Problem-09

Let t : **S**→[0,1] be a continuous or "fuzzy" truth function on the set **S** of statements. Define the implication operator as the truth function t_L(A→B)=min(1, 1-t(A)+t(B)) for statements A and B. Then prove the following generalized fuzzy *modus ponens* inference rule:

	t_L(A→B)	=	c
	t(A)	≥	α
Therefore	t(B)	≥	max(0, α+c-1)

Hence, if t(A)= t_L(A→B)= 1, then t(B)=1, which generalizes classical bivalent *modus ponens*.

Figure 8: data

- if $c \leq 1$
  then $1 - t(A) + t(B) = c \implies t(B) = c + t(A) - 1$
  From (2) $t(A) \geq a \implies t(A) + c - 1 \geq a + c - 1 \implies t(B) \geq c + a - 1$

  The value of $t(B)$ has to be between (0,1), so the general is : $t(B) \geq max(0, c + a - 1)($

- if $c = 1$
  General modus ponens.
  When $t(A) = 1$ and $t_L(A- > B) = min(1, 1 - t(A) + t(B))$ we have
  $t_L(A- > B) = min(1, 1 - 1 + t(B)) = min(1, t(B)) = 1$
  So $min(1, t(B)) = 1$ thus $t(B) = 1$

- Problem 10

$t_1(A- > B) = 1 = c$, (1)
$t(B) = 0 = b$, (2)
$t(A) <= min(1, 1 - c + b)$, (3)
From (1),(2) and (3), $t(A) = min(1, 1 - 1 + 0) = min(1, 0) = 0$

- Problem 11

$t(A\text{AND}\neg A) = min(t(A), t(\neg A)) = min(t(A), 1-t(A))$ $t(A\text{OR}\neg A) = max(t(A), t(\neg A)) = max(t(A), 1 - t(A))$
So $min(t(A), 1 - t(A)) + max(t(A), 1 - t(A))$ (1)
It is true although that t(A) is between (0,1) for example

- t(A) = 0.1
  (1) $\implies min(0.1, 0.9) + max(0.1, 0.9) = 1$

- t(A) = 0.5
  (1) $\implies min(0.5, 0.5) + max(0.5, 0.5) = 1$



Problem-11

Lukasiewicz's multivalued or fuzzy logic ($\mathbf{L_1}$ logic) uses a continuous-valued truth function et t : **S**→[0,1] defined on the set **S** of statements. Lukasiewicz defined the generalized conjunction (AND), disjunction (OR), negation (NOT) operators respectively as:

$$t(A \text{ AND } B) = \min(t(A), t(B))$$
$$t(A \text{ OR } B) = \max(t(A), t(B))$$
$$t(\text{NOT-}A) = 1 - t(A)$$

for statements A and B. Prove the generalized noncontradiction-excluded middle law:

$$t(A \text{ AND } \sim A) + t(A \text{ OR } \sim A) = 1.$$

Figure 9: data

- t(A) = 0.95
  (1) $\implies min(0.95, 0.05) + max(0.95, 0.05) = 1$
  So $min(t(A), 1 - t(A)) + max(t(A), 1 - t(A)) = 1$